

Homework5

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1 Question1

Minimize

$$J = \|\mathbf{x} - \hat{\mathbf{x}}\|^2 = \left\| \mathbf{x} - \sum_{i=1}^M \alpha_i \mathbf{e}_i \right\|^2$$

Also,

$$J = \left(\mathbf{x} - \sum_{i=1}^M \alpha_i \mathbf{e}_i \right)^T \left(\mathbf{x} - \sum_{j=1}^M \alpha_j \mathbf{e}_j \right) = \mathbf{x}^T \mathbf{x} - 2 \sum_{i=1}^M \alpha_i \mathbf{e}_i^T \mathbf{x} + \sum_{i=1}^M \sum_{j=1}^M \alpha_i \alpha_j \mathbf{e}_i^T \mathbf{e}_j.$$

The $\mathbf{e}_i$'s are orthonormal, $\mathbf{e}_i^T \mathbf{e}_j = \delta_{ij}$
we get:

$$J = \mathbf{x}^T \mathbf{x} - 2 \sum_{i=1}^M \alpha_i \mathbf{e}_i^T \mathbf{x} + \sum_{i=1}^M \alpha_i^2.$$

Taking the partial derivative and setting it to zero:

$$\frac{\partial J}{\partial \alpha_k} = -2 \mathbf{e}_k^T \mathbf{x} + 2\alpha_k = 0 \implies \alpha_k = \mathbf{e}_k^T \mathbf{x}, \quad k = 1, 2, \dots, M.$$

This is a minimum since $\frac{\partial^2 J}{\partial \alpha_k^2} = 2 > 0$

$\alpha_i = \mathbf{e}_i^T \mathbf{x}$ is the projection of the data vector $\mathbf{x}$ onto the eigenvector $\mathbf{e}_i$

2 Question2

The induced one-dimensional means and variances:

$$\mu_i = \mathbf{w}^T \boldsymbol{\mu}_i, \quad \sigma_i^2 = \mathbf{w}^T \boldsymbol{\Sigma}_i \mathbf{w}$$

(a) Maximizing $J_1(\mathbf{w})$

$$J_1(\mathbf{w}) = \frac{(\mu_1 - \mu_2)^2}{\sigma_1^2 + \sigma_2^2} = \frac{[\mathbf{w}^T(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)]^2}{\mathbf{w}^T(\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)\mathbf{w}} = \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_W \mathbf{w}},$$

with definition:

$$\mathbf{S}_B(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T, \quad \mathbf{S}_W \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2.$$

setting the gradient to zero:

$$\nabla_{\mathbf{w}} J_1 = \frac{2 \mathbf{S}_B \mathbf{w} (\mathbf{w}^T \mathbf{S}_W \mathbf{w}) - 2 \mathbf{S}_W \mathbf{w} (\mathbf{w}^T \mathbf{S}_B \mathbf{w})}{(\mathbf{w}^T \mathbf{S}_W \mathbf{w})^2} = \mathbf{0}$$

Also,

$$\mathbf{S}_B \mathbf{w} = \lambda \mathbf{S}_W \mathbf{w}, \quad \lambda = \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_W \mathbf{w}} = J_1(\mathbf{w}).$$

Considering that: $\mathbf{S}_B \mathbf{w} = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \underbrace{(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \mathbf{w}}_{\text{scalar}}$

Hence

$$\mathbf{S}_W \mathbf{w} \propto (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \implies \mathbf{w} \propto \mathbf{S}_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2),$$

since the scale of $\mathbf{w}$ is irrelevant

$$\mathbf{w} = (\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2).$$

(b) Maximizing $J_2(\mathbf{w})$

The argument is identical $\sigma_i^2 = \mathbf{w}^T \boldsymbol{\Sigma}_i \mathbf{w}$:

$$J_2(\mathbf{w}) = \frac{[\mathbf{w}^T(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)]^2}{\mathbf{w}^T [P(\omega_1)\boldsymbol{\Sigma}_1 + P(\omega_2)\boldsymbol{\Sigma}_2] \mathbf{w}} = \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_W \mathbf{w}}, \quad \mathbf{S}_W = P(\omega_1)\boldsymbol{\Sigma}_1 + P(\omega_2)\boldsymbol{\Sigma}_2.$$

By exactly the same reasoning as in (a), the maximizer satisfies $\mathbf{S}_W \mathbf{w} \propto (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$, so

$$\mathbf{w} = [P(\omega_1)\boldsymbol{\Sigma}_1 + P(\omega_2)\boldsymbol{\Sigma}_2]^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2).$$

(c) Comparison

maximizes

$$J_F(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_W \mathbf{w}}, \quad \mathbf{S}_W = \sum_{i=1}^2 \sum_{\mathbf{x} \in \mathcal{D}_i} (\mathbf{x} - \mathbf{m}_i)(\mathbf{x} - \mathbf{m}_i)^T,$$

Since $n_i/n \approx P(\omega_i)$ and each class scatter is n_i times the sample covariance, we have $\mathbf{S}_W \propto P(\omega_1)\boldsymbol{\Sigma}_1 + P(\omega_2)\boldsymbol{\Sigma}_2$.

Therefore, $J_2(\mathbf{w})$ is closer to criterion: its denominator is the prior weighted sum of class covariances. In contrast, $J_1(\mathbf{w})$ weights both classes equally regardless of prior probability, which corresponds to Fisher's LDA only in the special case $P(\omega_1) = P(\omega_2) = 1/2$.

3 Question3

3.1 (a)

Download the AReM data.

3.2 (b)

Train / Test Split.

3.3 (ci)time-domain

Including min, max, mean, median, std, Q1, Q3 (7 features * 6 time series = 42 features per instance)

3.4 (cii)feature extraction

Something wrong existed in the cycling/dataset9.csv and cycling/dataset14.csv.

Error tokenizing data. C error: Expected 7 fields in line 485, saw 8
File: /content/drive/MyDrive/USC_Summer26/EE559/Data/AReM/cycling/dataset14.csv

Error tokenizing data. C error: Expected 7 fields in line 485, saw 8
File: /content/drive/MyDrive/USC_Summer26/EE559/Data/AReM/cycling/dataset9.csv

Total bad files: 2

Results:

Time Series 1: avg_rss12							
Instance	min	max	mean	median	std	Q1	Q3
0	35.00	47.40	43.9545	44.33	1.5588	43.00	45.00
1	33.00	47.75	42.1798	43.50	3.6707	39.15	45.00
2	33.00	45.75	41.6781	41.75	2.2435	41.33	42.75
3	37.00	48.00	43.4550	43.25	1.3861	42.50	45.00
4	36.25	48.00	43.9691	44.50	1.6184	43.31	44.67

Time Series 2: var_rss12							
Instance	min	max	mean	median	std	Q1	Q3
0	0.0	1.70	0.4263	0.47	0.3387	0.00	0.50
1	0.0	3.00	0.6960	0.50	0.6309	0.00	1.12
2	0.0	2.83	0.5360	0.50	0.4055	0.43	0.71
3	0.0	1.58	0.3781	0.47	0.3156	0.00	0.50
4	0.0	1.50	0.4131	0.47	0.2631	0.43	0.50

Time Series 3: avg_rss13

Instance	min	max	mean	median	std	Q1	Q3
0	6.50	29.75	22.1224	23.000	3.0309	19.75	24.0000
1	8.50	30.00	22.1836	23.000	3.8105	20.50	24.3725
2	3.00	28.25	19.0066	19.125	4.0871	16.50	22.0625
3	5.75	27.00	15.7933	15.000	3.8476	13.00	18.2700
4	1.50	26.33	15.8680	16.250	3.7424	14.25	18.0000

Time Series 4: var_rss13

Instance	min	max	mean	median	std	Q1	Q3
0	0.0	4.44	0.4973	0.43	0.5507	0.00	0.83
1	0.0	5.15	0.9899	0.83	0.9537	0.43	1.30
2	0.0	6.42	0.8419	0.50	0.9288	0.43	1.12
3	0.0	10.03	0.8494	0.50	0.9958	0.43	1.12
4	0.0	5.17	0.6664	0.47	0.7890	0.00	0.94

Time Series 5: avg_rss23

Instance	min	max	mean	median	std	Q1	Q3
0	29.00	38.50	35.5885	36.00	1.9996	35.3625	36.50
1	20.00	38.67	33.4939	35.00	3.8494	30.4575	36.33
2	23.67	37.50	29.8571	30.00	2.4110	28.4575	31.25
3	8.00	33.50	23.0348	23.50	2.4889	22.2500	24.00
4	11.33	30.75	22.1038	21.67	3.3183	20.5000	23.75

Time Series 6: var_rss23

Instance	min	max	mean	median	std	Q1	Q3	label
0	0.0	1.79	0.4933	0.43	0.5135	0.00	0.94	bending1
1	0.0	2.18	0.6135	0.50	0.5243	0.00	1.00	bending1
2	0.0	1.79	0.3833	0.43	0.3892	0.00	0.50	bending1
3	0.0	5.26	0.6796	0.50	0.6225	0.43	0.87	bending1
4	0.0	2.96	0.5553	0.49	0.4878	0.00	0.83	bending1

5 rows x 43 columns

3.5 (ciii)Bootstrap Confidence

Results shown in the table below:

Feature	Std. Dev.	CI Lower (5%)	CI Upper (95%)
min_avg_rss12	8.8564	7.5743	10.1311
max_avg_rss12	4.1920	3.0131	5.1613
mean_avg_rss12	4.8254	4.2452	5.3171
median_avg_rss12	4.8730	4.2282	5.4061
std_avg_rss12	1.7571	1.5399	1.9315
q1_avg_rss12	5.6901	5.1159	6.1642
q3_avg_rss12	4.6223	3.7824	5.3478
min_var_rss12	0.0000	0.0000	0.0000
max_var_rss12	5.1442	4.6426	5.5194
mean_var_rss12	1.6052	1.4047	1.7425
median_var_rss12	1.4383	1.2442	1.5764
std_var_rss12	0.9050	0.8154	0.9656
q1_var_rss12	0.9545	0.8263	1.0481
q3_var_rss12	2.1655	1.9065	2.3396
min_avg_rss13	3.0523	2.8254	3.2216
max_avg_rss13	4.6877	3.9136	5.3259
mean_avg_rss13	3.8213	3.1458	4.3752
median_avg_rss13	3.8094	3.1158	4.3806
std_avg_rss13	1.0031	0.7880	1.1960
q1_avg_rss13	4.0999	3.4174	4.6510
q3_avg_rss13	3.9196	3.2164	4.5032
min_var_rss13	0.0000	0.0000	0.0000
max_var_rss13	2.3023	2.0655	2.4877
mean_var_rss13	1.1827	1.0823	1.2398
median_var_rss13	1.1534	1.0517	1.2087
std_var_rss13	0.4743	0.4313	0.5057
q1_var_rss13	0.8442	0.7660	0.8956
q3_var_rss13	1.5701	1.4354	1.6475
min_avg_rss23	5.3433	3.3536	6.9368
max_avg_rss23	5.4893	4.2958	6.4153
mean_avg_rss23	5.1498	3.6282	6.3434
median_avg_rss23	5.2998	3.6831	6.5568
std_avg_rss23	1.0424	0.7775	1.2652
q1_avg_rss23	5.5681	4.0234	6.7848
q3_avg_rss23	4.9930	3.5932	6.1568
min_var_rss23	0.0521	0.0000	0.0890
max_var_rss23	2.5590	2.2430	2.8452
mean_var_rss23	1.1743	1.0700	1.2312
median_var_rss23	1.1088	1.0062	1.1668
std_var_rss23	0.5186	0.4761	0.5463
q1_var_rss23	0.7780	0.7035	0.8261
q3_var_rss23	1.5545	1.4197	1.6270

3.6 (civ)Selection

Feature	Separation Score
max_avg_rss23	2.4866
mean_avg_rss23	1.9848
q3_avg_rss23	1.9647
median_avg_rss23	1.8711
q1_avg_rss23	1.8163
min_avg_rss23	1.7494

Six of the top six features all come from the same time series, **avg_rss23**, strongly s is the most discriminative for distinguishing bending from other activities.

Then, among the six **avg_rss23** features, **min**, **mean**, and **max** were chosen.

3.7 (cv)Plots

Results are shown below:

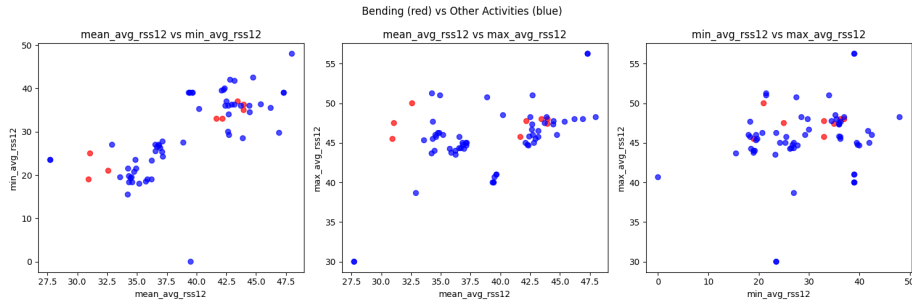


Figure 1: Scatter plots of **mean_avg_rss12**, **min_avg_rss12**, and **max_avg_rss12** for the training instances, comparing bending activities (red) against all other activities (blue).

4 Question4

4(a)i

The scatter plots of the selected features (mean, min, max) extracted from both halves of time series 1, 2, and 6, with bending in red and other activities in blue.

Discussion. The half-series feature clouds look essentially the same as the full-series ones from 3(c)v. The series are roughly stationary within an instance, so splitting does not create considerably different separability; it just doubles the number of correlated features.

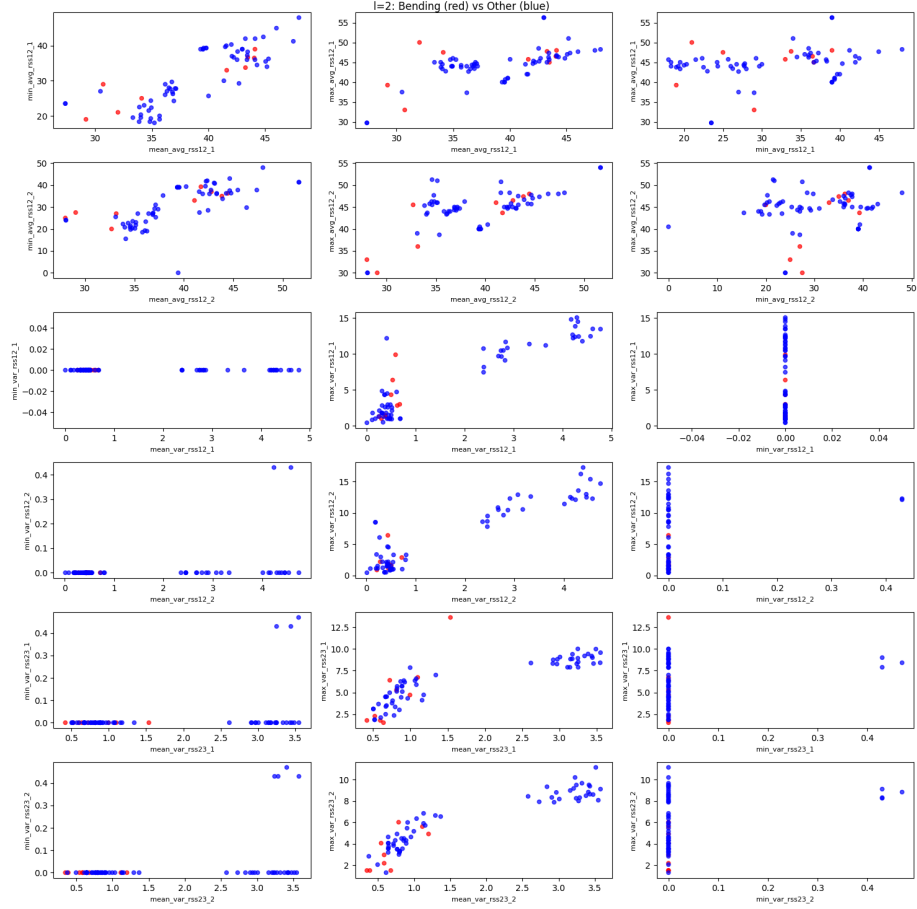


Figure 2: $l = 2$: scatter plots of the selected features extracted from both halves of time series 1, 2, and 6. Bending (red) vs. other activities (blue).

4(a)ii

Features were selected by RFE around an unregularized ($C \rightarrow \infty$) logistic regression, and the pair was chosen by stratified 5-fold cross-validation.

Stratified CV. The training set is imbalanced (9 bending vs. 60 other), so plain folds could contain no bending instances. Stratified 5-fold CV preserves the class proportion in every fold.

Result.

$$(l^*, p^*) = (1, 23), \quad \text{CV accuracy} = 0.9857.$$

$l = 1$ winning is consistent with 4(a)i

the series are roughly stationary, so larger l only adds redundant correlated features.

4(a)iii

The final model ($l = 1, p = 23$) achieves perfect classification on the training set:

$$\text{Train confusion matrix} = \begin{pmatrix} 60 & 0 \\ 0 & 9 \end{pmatrix}, \quad \text{AUC} = 1.000.$$

The training ROC curve is shown in Figure 3.

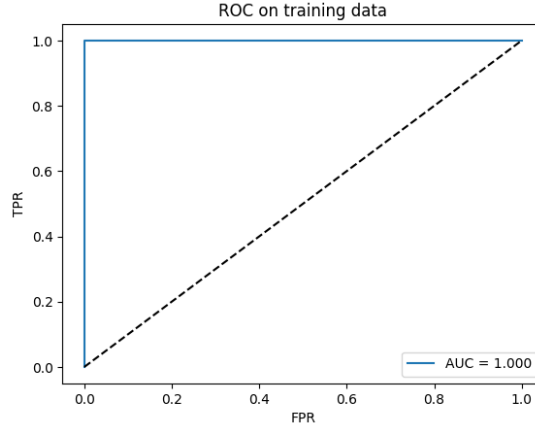


Figure 3: ROC curve on the training data (AUC = 1.000).

4(a)iv

The test series were broken into the same $l = 1$ segments chosen by CV:

$$\text{Test accuracy} = 1.0000 \quad \text{vs.} \quad \text{CV accuracy} = 0.9857, \quad \text{Test confusion matrix} = \begin{pmatrix} 15 & 0 \\ 0 & 4 \end{pmatrix}.$$

Feature	β	p-value
intercept	-10.4401	—
min_avg_rss12	-1.8064	—
max_avg_rss12	1.2482	—
mean_avg_rss12	-1.1429	—
median_avg_rss12	-0.5413	—
q3_avg_rss12	-2.3771	—
max_var_rss12	-1.2251	—
max_avg_rss13	1.5092	—
median_avg_rss13	-0.7302	—
std_avg_rss13	-1.3330	—
q3_avg_rss13	-1.3213	—
max_var_rss13	1.8160	—
median_var_rss13	-0.7894	—
q1_var_rss13	-0.8792	—
min_avg_rss23	3.9271	—
max_avg_rss23	4.3959	—
mean_avg_rss23	1.7922	—
q1_avg_rss23	1.3447	—
q3_avg_rss23	0.9543	—
mean_var_rss23	-0.6886	—
median_var_rss23	-0.7205	—
std_var_rss23	-1.1766	—
q1_var_rss23	-1.4584	—
q3_var_rss23	-0.6783	—

Table 1: Logistic regression coefficients ($l = 1$, $p = 23$, standardized features). P-values are unavailable due to perfect separation.

All 19 test instances are classified correctly. The two accuracies are consistent (the CV estimate reflects a single misclassified instance), confirming the model generalizes well.

4(a)v

Yes.

The classes are linearly separable: training and test classification are both perfect, and the unpenalized ML fit raised a *perfect separation* error, so the p-values are undefined.

The `sklearn` coefficients stay finite ($\max |\beta| = 4.40$) only because of the residual $\mathcal{L}_2$ penalty at $C = 10^6$.

4(a)vi

Yes

the classes are imbalanced: 9 bending with 60 other in training.

A case-control model:

$$\beta'_0 = \beta_0 + \log \frac{\pi_1}{\pi_0} - \log \frac{\tilde{\pi}_1}{\tilde{\pi}_0},$$

where $\pi_1 = 9/69$ is the true bending prior and $\tilde{\pi}_1 = 0.5$ the sampling proportion. On the test set:

$$\text{Confusion matrix} = \begin{pmatrix} 14 & 1 \\ 0 & 4 \end{pmatrix}, \quad \text{AUC} = 0.983.$$

The ROC curve is shown in Figure 4. Performance is close to the original model.

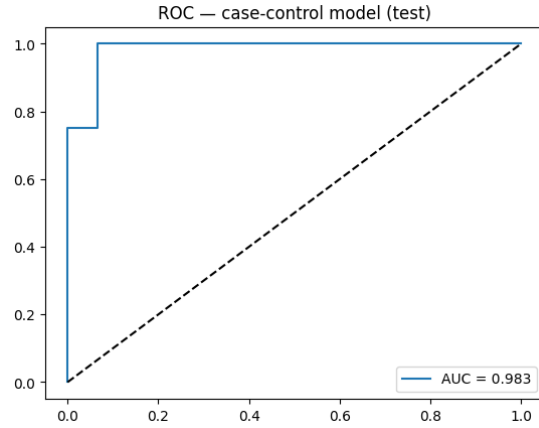


Figure 4: ROC of the case-control model on the test set (AUC = 0.983).

4(b)i

Features were standardized and `LogisticRegressionCV` cross-validated λ automatically for each l , so both l and λ are cross validated.

Best configuration:

$$l^* = 1, \quad C^* \approx 3.79 \times 10^3, \quad \text{CV accuracy} = 0.9857, \quad \text{Test accuracy} = 1.0000,$$

4(b)ii

Performance: The two approaches perform equally well here: both reach CV accuracy 0.9857 and test accuracy 1.0000.

But $\mathcal{L}_1$ is easier: it gets everything done in one step and is stable.

The p-value method involves more steps and its theoretical assumptions are violated by this dataset

Therefore, $\mathcal{L}_1$ is preferred.

4(c)i

The best l was found as in 4(b)i with $\mathcal{L}_1$ -penalized multinomial regression; λ is cross-validated automatically.

Best test error = 0.1579.

Confusion matrix generalizes to a 7×7 table, and ROC curves are defined one-vs-rest.

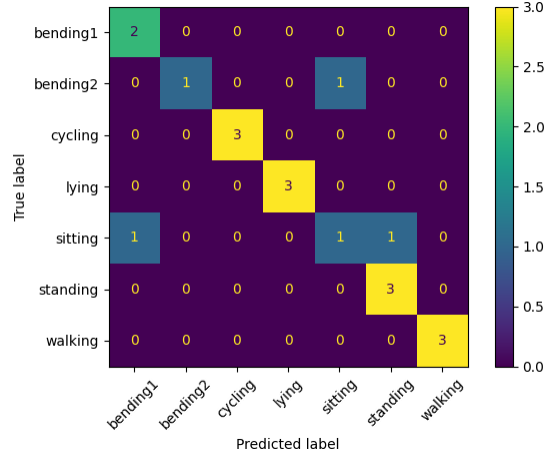


Figure 5: Test confusion matrix — multinomial $\mathcal{L}_1$.

4(c)ii

Gaussian NB better than the Multinomial NB. As l grows added segments are correlated, violating the independence assumption.

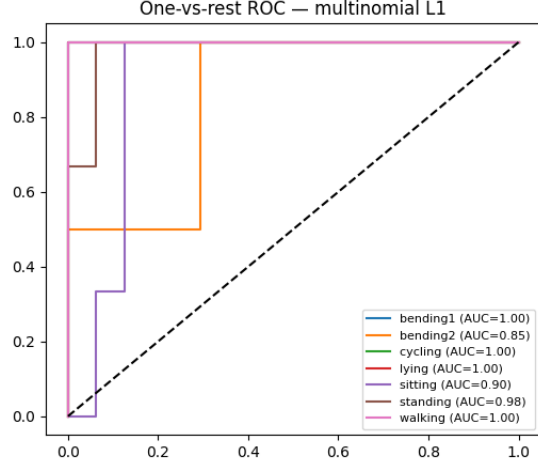


Figure 6: One-vs-rest ROC curves (test) with per-class AUCs.

l	Gauss.	Multin.	l	Gauss.	Multin.
1	0.8571	0.7231	11	0.6824	0.6659
2	0.8275	0.7516	12	0.5945	0.6659
3	0.6813	0.7231	13	0.6110	0.6802
4	0.7253	0.7231	14	0.5538	0.6802
5	0.6527	0.7231	15	0.5659	0.6659
6	0.6527	0.6945	16	0.5538	0.6659
7	0.6956	0.6945	17	0.5670	0.6516
8	0.6670	0.7231	18	0.4802	0.6802
9	0.6385	0.6945	19	0.5385	0.6802
10	0.6813	0.6802	20	0.5374	0.6516

Table 2: 5-fold CV accuracy for each l (best per column in bold). GaussianNB: $l^* = 1$, test error = 0.1053; MultinomialNB: $l^* = 2$, test error = 0.2105.

4(c)iii

PCA was placed inside the CV pipeline, grid-searching (l, p) :

$$(l^*, p^*) = (1, 10), \quad \text{CV accuracy} = 0.9143, \quad \text{test error} = 0.1579.$$

Errors concentrate in the static postures (Figure 7); the PC1–PC2 scatter (Figure 8) shows cycling, walking, and lying separating cleanly while bending2/sitting/standing overlap.

PCA decorrelates the features, suiting Gaussian NB’s independence assumption (CV accuracy $0.857 \rightarrow 0.914$).

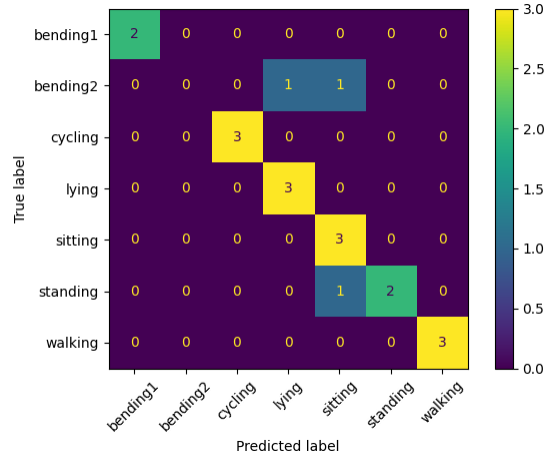


Figure 7: Test confusion matrix — PCA + Gaussian NB, $(l, p) = (1, 10)$.

4(c)iv

Method	Test error
Multinomial $\mathcal{L}_1$ logistic	0.1579
GaussianNB	0.1053
MultinomialNB	0.2105
PCA + GaussianNB	0.1579

Table 3: Test errors of the multiclass methods.

On this dataset, Gaussian Naïve Bayes achieves the lowest test error (0.1053), followed by multinomial $\mathcal{L}_1$ logistic regression and PCA + Gaussian NB.



Figure 8: Training data on the first two PCs ($l = 1$).